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The auditory dynamic attending theory revisited: A closer look at the pitch comparison task

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ABSTRACT

The dynamic attending theory as originally proposed by Jones, 1976. Psychol. Rev. 83(5), 323–355 posits that tone sequences presented at a regular rhythm entrain attentional oscillations and thereby facilitate the processing of sounds presented in phase with this rhythm. The increased interest in neural correlates of dynamic attending requires robust behavioral indicators of the phenomenon. Here we aimed to replicate and complement the most prominent experimental implementation of dynamic attending (Jones et al., 2002. Psychol. Sci. 13(4), 313–319). The paradigm uses a pitch comparison task in which two tones, the initial and the last of a longer series, have to be compared. In-between the two, distractor tones with variable pitch are presented, at a regular pace. A comparison tone presented in phase with the entrained rhythm is hypothesized to lead to better behavioral performance. Aiming for a conceptual replication, four different variations of the original paradigm were created which were followed by an exact replication attempt. Across all five experiments, only 40 of the 140 tested participants showed the hypothesized pattern of an inverted U-shaped profile in task accuracy, and the group average effects did not replicate the pattern reported by Jones et al., 2002. Psychol. Sci. 13(4), 313–319 in any of the five experiments. However, clear evidence for a relationship between musicality and overall behavioral performance was found. This study casts doubt on the suitability of the pitch comparison task for demonstrating auditory dynamic attending. We discuss alternative tasks that have been shown to support dynamic attending theory, thus lending themselves more readily to studying its neural correlates.

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1. Introduction

Many sounds in our environment can be characterized to some degree by temporal regularity or periodicity, such as speech or music. Perceiving regularities is beneficial for understanding the acoustic information (Arnal and Giraud, 2012) and for increasing its perceptual coherence so that it stands out from other concurrent information (for a recent overview see Bendixen, 2014; Winkler et al., 2009). Temporally regular (i.e., rhythmic) stimulation has been shown to induce strong temporal expectations for upcoming events (Jones, 2010; Mathewson et al., 2010; Schroeder and Lakatos, 2009). Such expectations are considered to be created exogenously when the input dynamics have a nonrandom temporal pattern (Nobre et al., 2007). Previous research investigating temporal expectations has found evidence for facilitated motor behavior (Sanabria et al., 2011) as well as improved discrimination ability (Rohenkohl et al., 2012) in response to temporally anticipated events. In the auditory modality, attention in time is reflected for instance in musical expectancies. Since auditory patterns unfold over time, the role of temporal expectancies as caused by stimulus timing can be considered crucial for auditory processing (Barnes and Jones, 2000). Indeed, various studies demonstrate that temporally expected sounds are preferentially processed (e.g., Lange and Röder, 2006; Lange et al., 2003; Lange, 2009).

The concept of preferential processing was first described by the auditory dynamic attending theory (Jones and Boltz, 1989; Jones, 1976), which predicts that tone sequences presented at a regular rhythm entrain attentional oscillations and thus facilitate the processing of sounds presented in phase with this rhythm (Jones et al., 2002; Large and Jones, 1999). The neural substrate underlying this preferred processing is thought to be the ongoing neural oscillations that can be entrained by rhythmic stimuli and thus align their temporal dynamics to external patterns (Calderone et al., 2014; Thorne et al., 2011; for a recent review see: Henry and Herrmann, 2014). The term entrainment in general relates to one oscillator falling into step with another. Whereas *sensory entrainment* refers to the regular presentation of a series of sensory stimuli that entrain for example attentional oscillations, *neural entrainment* on the other hand relates to neural oscillations falling into step with a sequence of temporally regular events, such as a series of sensory stimuli (e.g., Henry and Obleser, 2012). As neural entrainment can optimize neural excitability to be high (or low) when a stimulus is expected (Arnal and Giraud, 2012; Lakatos et al., 2008, 2013), this mechanism can explain why predictable stimuli are more easily perceived than random, unpredictable stimuli (Barnes and Jones, 2000; Jones et al., 2002; Large and Jones, 1999; Mathewson et al., 2010; Rohenkohl et al., 2012). Interestingly, entrainment has also been proposed to underlie selective attention in the context of multiple concurrent stimulus streams (Lakatos et al., 2013; Schroeder and Lakatos, 2009).

Partly driven by uncovering such neural correlates and mechanisms of temporal attention, the dynamic attending theory (Jones, 1976) has experienced an upsurge in research interest and activity. One particular experimental implementation of dynamic attending by Jones et al. (2002) has become widely popular both in neuroscience and music psychology research during the past decade. In neuroscience the conceptual fit of the dynamic attending theory framework to the idea of neural entrainment has been observed, as stimulus events occurring at excitable phases of entrained neural oscillations are for example more likely to be detected or be responded to more quickly (Cravo et al., 2013; Henry and Herrmann, 2014; Schroeder and Lakatos, 2009; Stefanics et al., 2010). Henry and Herrmann (2014) further suggested that low-frequency oscillations in the delta-theta range might reflect correlates of the attentional oscillation, and that the excitable phase of neural oscillations might correspond to the peak of the attentional pulse in the dynamic attending theory framework.

Music psychology research on the other hand has used the idea of dynamic attending as the basis of an experimental approach to investigate the impact of rhythmicity on auditory processing. Accordingly, the Jones et al. (2002) paper has been widely cited (163 citations in Web of Knowledge and 251 citations in Google Scholar as of March, 2015). This suggests that the paradigm described therein gives a promising research tool for further studies on dynamic attending and its neural correlates. In the present study, we set out to test the robustness of this paradigm.

The original paradigm by Jones et al. (2002), which is illustrated in Fig. 1A, is a pitch comparison task, where an initial tone (standard) has to be compared to a final tone (comparison), presented at the same, higher or lower pitch level (not illustrated in Fig. 1A). Between the standard and comparison tones, a series of eight intervening tones (distractors) is presented with a constant stimulus-onset-asynchrony (SOA) of 600 ms, evoking a temporal expectancy. Importantly, the comparison tone can be presented either at the expected SOA (i.e., in phase with the distractor sequence), or earlier or later, thus out of phase (for a detailed description see Section 4.3). The dynamic attending theory predicts that task performance is on average better in trials when the comparison tone is in phase with the presented rhythm, as compared to either type of out-of-phase presentation (early or late). This leads to an inverted U-shaped expectancy profile describing the dependence of task performance on presentation time of the comparison tone. Further, the results were extended by Jones et al. (2002) in a second experiment by showing that the effect of an inverted U-shaped profile could be extrapolated into the future by inserting a silent cycle after the last distractor appearance.

However, in the study of Jones et al. (2002), the standard pitch was repeated once as the final distractor tone (for illustration see Fig. 1A). According to Jones et al. (2002) this manipulation was aimed at making the task less difficult, thus boosting task performance and additionally preventing spurious biasing from the pitch difference between the last distractor and the comparison tone. In our view this repetition raises important methodological concerns. The pitch-comparison task requires the pitch of the standard tone to be stored in memory, while the pitches of the intervening sequence are ignored, after which the comparison tone is then compared to the standard tone. Yet, if the last distractor tone is identical to the standard tone, there is no need for memory storage or even attention to the standard and consequently no need to suppress the intervening tones. Further, if the last distractor is beneficial for task performance it

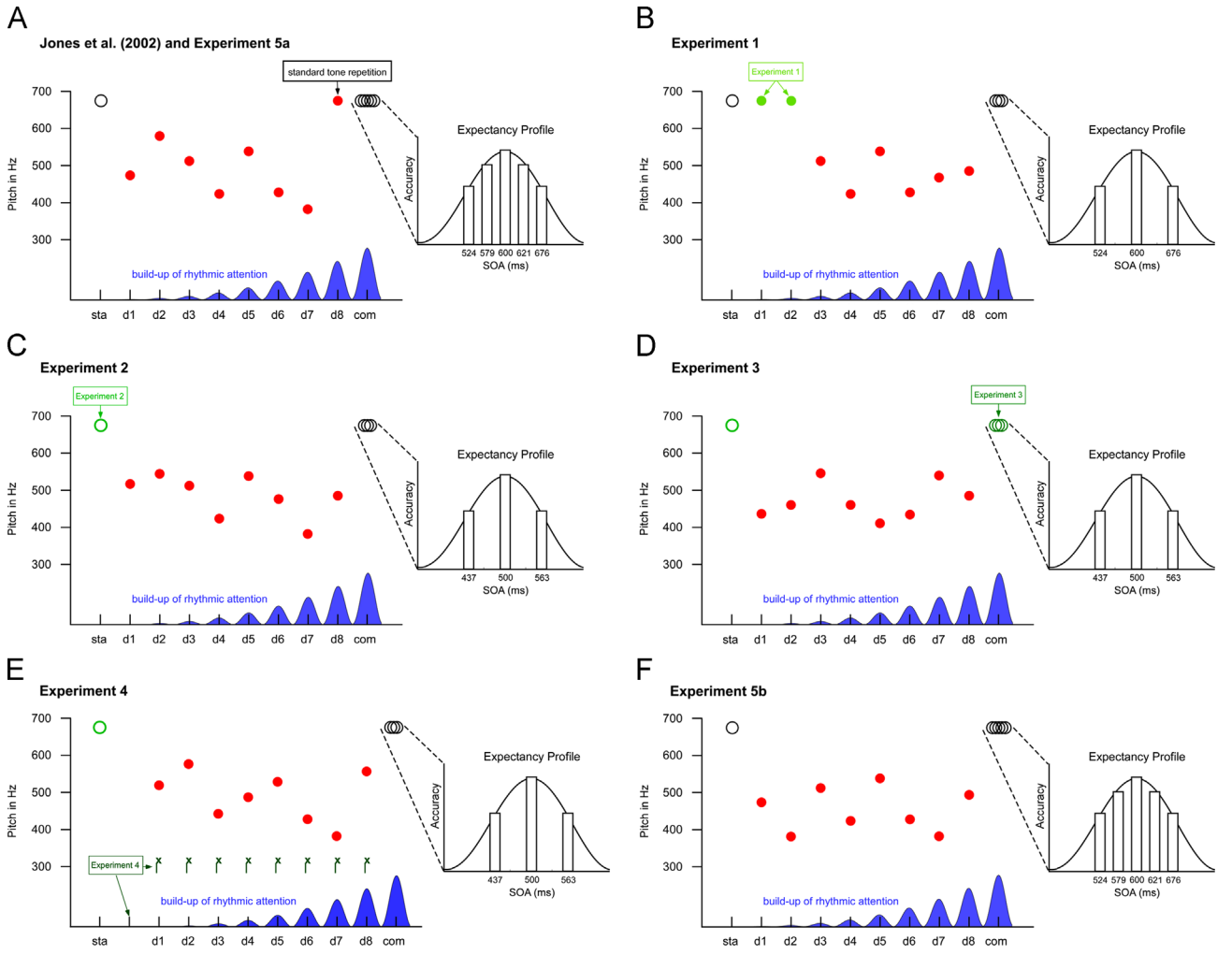


Fig. 1 – Schematic illustration of the pitch comparison paradigm. In all versions of the task, an initial standard tone (sta) is presented, followed by 8 distractors (d1–d8), and a final comparison tone (com). The task is to judge the pitch of the comparison in relation to the standard tone. The regular presentation of the distractor sequence is thought to gradually build-up rhythmic attention, oscillating at the entrainment frequency. (A) Original paradigm conducted by Jones et al. (2002) and implemented in Experiment 5a. Modifications implemented in the present experiments are illustrated in green. (B) Experiment 1: repetition of the standard tone as the first two distractor tones (C) Experiment 2: individual threshold of standard tone, 100 ms to 250 ms, in steps of 25 ms as determined by an adaptive tracking procedure (D) Experiment 3: individual threshold of standard tone and shortening of the comparison tone (100 ms) (E) Experiment 4: individual threshold of standard tone; further, a silent cycle was included between the standard and the first distractor; distractor tones were underlined with metronome clicks. (F) Experiment 5: using a within-subject design the original paradigm with the standard tone repetition was implemented (A, condition 1) whereas in condition 2 (F) the standard tone was not repeated. Critical SOAs between the last distractor tone and the comparison tone were 524 ms, 600 ms, and 676 ms in Experiment 1, 437 ms, 500 ms and 563 ms in Experiments 2 to 4, and 524 ms, 579 ms, 600 ms, 621 ms, and 676 ms in Experiment 5, respectively. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

is in fact no longer a distractor but a task-relevant tone. An initial pilot study ($N=4$) performed in our lab, exactly replicating the study by Jones et al. (2002), confirmed this notion. Thus, 50% of our participants reported comparing the last distractor tone (rather than the initial standard tone) with the comparison tone (Hagen, 2013). While participants recognizing this option were discarded from the statistical analysis in the study of Jones et al. (2002), it remains unclear to what extent the remaining participants might have implicitly used the information of a repeated standard tone for task processing. It would be reassuring to see the inverted U-shaped profile without repetition of the standard tone in order to assess the generalizability of this paradigm for studying attentional entrainment by rhythmic stimuli.

This methodological concern, as well as our aim to transfer the paradigm to a version suitable for measurement of neural oscillations, has served as the starting point for the present study. Specifically, the current study aimed to replicate Jones' initial findings by using a conceptual rather than exact replication for two reasons. First, as mentioned above we had concerns about the original design and second,

conceptual replications are essential for establishing generalizability of a scientific theory. Indeed, while exact replications of experiments are more likely to detect methodological concerns, a sufficiently large amount of conceptual replications serve this function as well (Frieler et al., 2013).

In a series of five experiments we investigated the effect of temporal expectation on task performance. In Experiment 1 we kept to the original procedures and methods as closely as possible, with the exception that the repeated standard pitch was used for the first two distractors instead of the last. By repeating the standard tone twice, we provided better sensory encoding of the standard tone. However, this manipulation might have reduced the attentional entrainment effect, as better sensory encoding could result in less reliance on contextual information. Thus, we excluded the repetition in Experiment 2.

Further, the standard tone duration was individualized by an adaptive tracking procedure in order to adjust task difficulty. In Experiment 3 we focused on manipulating the comparison tone so as to increase the chances of finding effects of temporal expectation. A more complex version of the paradigm was established in Experiment 4 including a silent cycle between the standard tone and the first distractor. Further, an underlying metronome was inserted for the distractors to give a more precise rhythmic structure. The metronome was additionally expected to reduce distractor interference, as it provides additional temporal information that is independent from pitch. To directly compare the influence of the repeated standard tone we conducted a final Experiment 5 where we implemented two conditions: an exact replication of the original paradigm and a condition without the repetition of the standard tone in a balanced within-subject design (for illustration see Fig. 1; for a detailed description of Experiments 1 to 5 see Section 4.4).

2. Results

In all five experiments, the group average accuracy was well above chance level (33%) with a mean proportion of correct responses in 56% of all trials (Experiment 1: 53%; Experiment 2: 58%; Experiment 3: 54%; Experiment 4: 68%; Experiment 5—Condition 1: 52%; Experiment 5—Condition 2: 50%). We found large inter-individual differences in task performance in all five experiments that varied between 33% and 97%. The overall accuracies are shown in Table 1 and individual accuracy profiles, separated for Experiments 1 to 5, are shown in Figs. 2 and 3, respectively.

As task performance in Experiment 1 was relatively poor, we implemented for Experiments 2 to 4 a two-down one-up adaptive tracking procedure, where the standard tone was adjusted individually (100 ms to 250 ms, in steps of 25 ms), to improve overall task performance.

The individual adjustment facilitated the task for Experiment 4 ($t(60) = -4.34$, $p < .001$), but not for Experiment 2 ($t(60) = -1.71$, $p = .09$) or Experiment 3 ($t(60) = -0.27$, $p = .79$). Task performance between the first training run at the start of the experiment and the second training run following the adaptive tracking procedure (see Section 4.4 for details) revealed a significant improvement (across Experiments 2 to 4: 51 to 58%; $t(65) = -2.89$, $p < .01$, $d = 0.32$). For around 55% of the participants, the adaptive tracking procedure resulted in the longest standard tone duration of 250 ms (Experiment 2: 12 of 22; Experiment 3: 17 of 22, Experiment 4: 7 of 22).

The main question was whether in-phase presentation of the comparison tone would result in better task performance than out-of-phase (i.e., early or late) presentation. For this we sought evidence of an inverted U-shaped response profile defined as higher percentage of correct response for in phase performance compared to out of phase performance (Experiments 1 to 4: very early < on time > very late; Experiment 5: very early < early < on time > late > very late). Over all five experiments, 40 out of the 140 participants showed the hypothesized inverted U-shaped profile at a descriptive level (Experiment 1: 17 of 40, Experiment 2: 7 of 22; Experiment 3: 5 of 22; Experiment 4: 9 of 22; Experiment 5—Condition 1: 1; Experiment 5—Condition 2: 1). To assess this observation statistically, a one-way repeated-measures ANOVA was performed on the group average for the Experiments 1 to 4 with accuracy as the dependent variable, employing the factor *timing of comparison tone* (very early/on time/very late). No significant effect of the temporal manipulation was found in any of the four experiments (Experiment 1: $F(2, 78) = 0.66$, $p = .52$, $\eta^2 = .017$; Experiment 2: $F(2, 42) = 0.59$, $p = .56$, $\eta^2 = .027$;

Table 1 – Overall accuracies for Experiments 1 to 5.

	Very early M (SD)	Early M (SD)	In phase M (SD)	Late M (SD)	Very late M (SD)
Experiment 1	.52 (.10)		.53 (.12)		.53 (.11)
Experiment 2	.57 (.15)		.58 (.14)		.59 (.15)
Experiment 3	.54 (.15)		.53 (.15)		.54 (.15)
Experiment 4	.67 (.17)		.68 (.17)		.68 (.16)
Experiment 5					
Condition 1	.48 (.12)	.54 (.12)	.51 (.13)	.53 (.12)	.52 (.12)
Condition 2	.51 (.11)	.51 (.09)	.50 (.19)	.50 (.10)	.49 (.11)

Note. Accuracy values are group means (M) and standard deviations (SD, in parentheses). For all reported accuracy values, chance level was 33%.

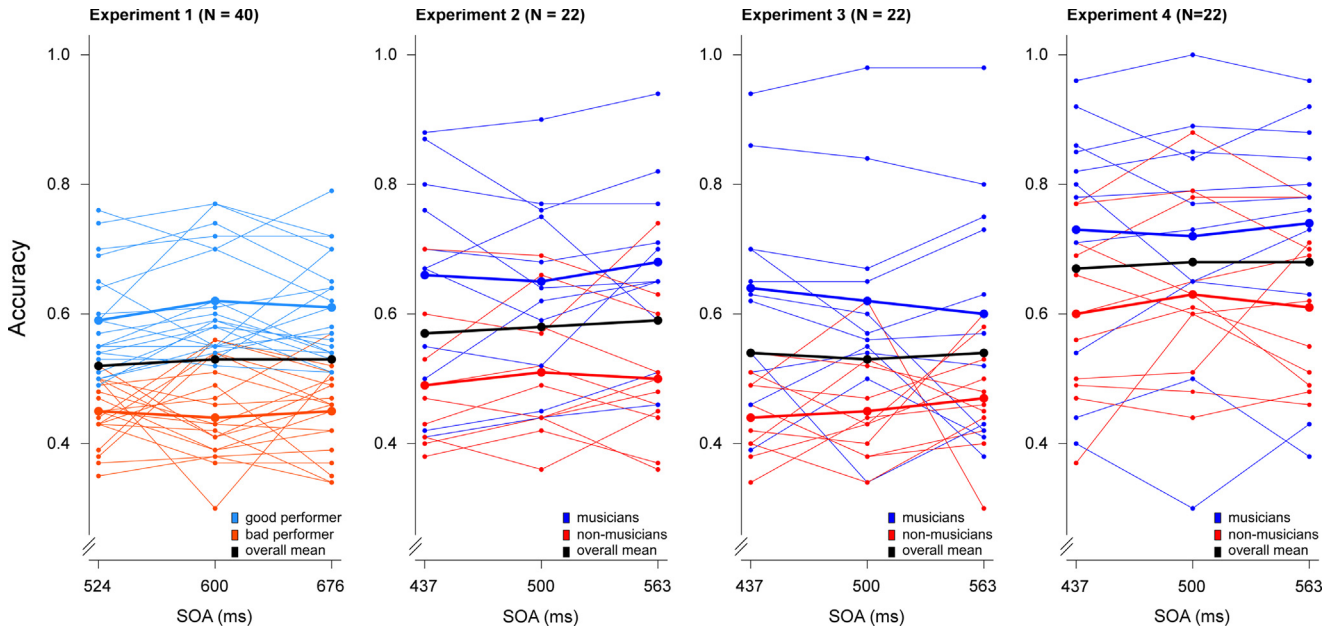


Fig. 2 – Individual accuracy profiles separately for Experiments 1 to 4. Bold lines refer to group means (black) and sub-group means, separately shown for good and bad performers (Experiment 1) and musicians versus non-musicians in Experiments 2 to 4. Note that chance level was 33%.

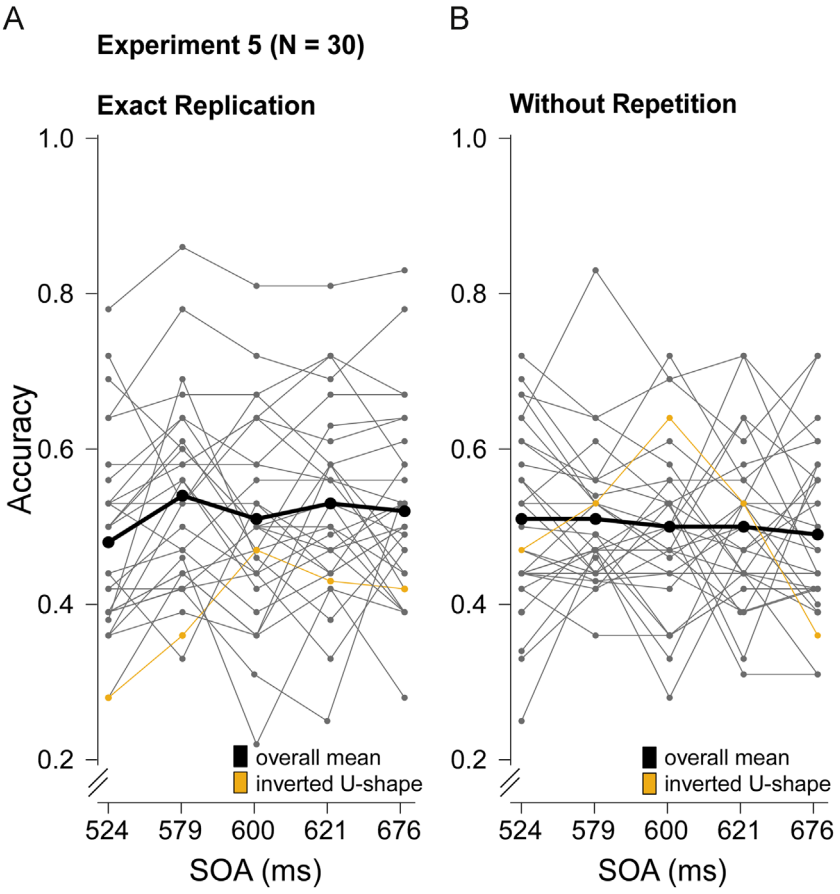


Fig. 3 – Individual accuracy profiles for Experiment 5, separated for the two conditions (A) with standard tone repetition and (B) without repetition. Bold lines refer to the group mean (black). Participants showing an inverted U-shaped profile are highlighted in yellow. Note that chance level was 33%. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

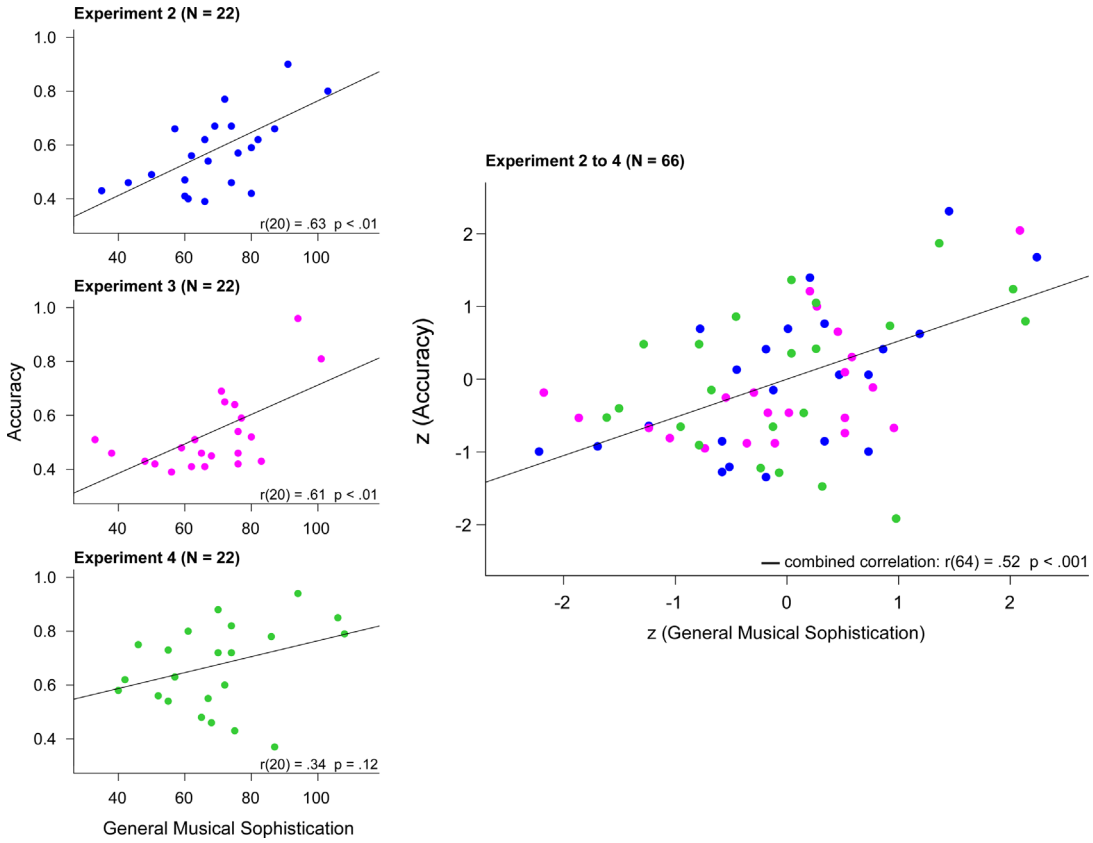


Fig. 4 – Systematic relationship between General Musical Sophistication and task accuracy for Experiments 2 to 4 separately (left) and pooled (right).

Experiment 3: $F(2, 42)=0.05$, $p=.95$, $\eta^2=.003$; Experiment 4: $F(2, 42)=0.22$, $p=.80$, $\eta^2=.010$). Note that all F values were considerably below 1, despite sufficient sample size and testing power ($1-\beta=.80$) determined by an a priori power analysis (see Section 4.5 for details).

For Experiment 5 (exact replication and condition without standard tone repetition) a 2×5 repeated-measures ANOVA was performed, with factors *timing of comparison tone* (very early/early/on time/late/very late) and *condition* (with and without repetition). The main effect of temporal manipulation was not significant ($F(4, 232)=1.17$, $p=.32$, $\eta^2=.020$), nor was the factor condition ($F(1, 58)=0.35$, $p=.56$, $\eta^2=.006$) or the interaction ($F(4, 232)=1.70$, $p=.15$, $\eta^2=.029$). Individual accuracy profiles separated for the condition with and without standard tone repetition are shown in Fig. 3.

Musicality in Experiments 2 to 4 ranged from non-musicians to well-trained musicians (General Musical Sophistication Factor: Experiment 2: range 35–110, $M=69$, $SD=15$; Experiment 3: range 33–101, $M=68$, $SD=16$; Experiment 4: range 40–108, $M=68$, $SD=19$). A Pearson product-moment correlation revealed a systematic relationship between task accuracy and musical sophistication in Experiments 2 ($r(20)=.63$, $p<.01$) and Experiment 3 ($r(20)=.61$, $p<.01$) which is illustrated in Fig. 4. Experiment 4 showed a relationship in the same direction that was not significant ($r(20)=.34$, $p=.12$). The overall correlation between musical sophistication and task performance across Experiments 2 to 4 was significant ($r(64)=.52$, $p<.001$; see Fig. 4).

A median split on the General Musical Sophistication Factor was used to categorize musicians versus non-musicians separately for each experiment. To assess whether musicians or non-musicians benefitted from temporal expectation in general, the difference between the mean accuracies for in-phase and out-of-phase (early and late) comparison tones was calculated. A subsequent post hoc analysis for Experiments 2 to 4 testing the difference between musicians and non-musicians regarding the influence of rhythmicity showed no significant results (Experiment 2: $t(20)=-1.194$, $p=.25$, $d=0.53$; Experiment 3: $t(20)=.183$, $p=.86$, $d=0.08$; Experiment 4: $t(20)=-1.340$, $p=.20$, $d=0.60$).

3. Discussion

The present study was designed to replicate the findings of Jones et al. (2002) in order to assess the robustness of this paradigm for investigating dynamic attending. Yet, no replication was achieved in any of the five experiments. Neither in the conceptual replications nor in the exact replication did we find behavioral evidence that participants performed better in a pitch comparison task at temporally expected events than at unexpected time intervals, despite an a priori power of $1-\beta=.80$. Even when separating participants according to musical expertise (Experiments 2 to 4), no task performance benefit by the rhythmic manipulation was seen in the musicians (who can be assumed to pick up the

rhythmicity of the sequence and detect deviations from it more easily, see Geiser et al. (2009)). Looking at the relationship between musicality and overall task accuracy showed however that musicians in general performed the task better than non-musicians—suggesting that the variance in musicality was large enough to detect inter-individual differences.

In order to understand our failure to replicate the effects of Jones et al. (2002), it is important to consider the factors that contribute to solving the task. First, the standard tone must be encoded and kept in mind (which might have put the musicians in Experiments 2 to 4 at an advantage, as musical sophistication includes perceptual abilities). Second, the pitch difference between the standard and comparison tone must be assessed. Since we used the same pitch differences as Jones et al. (2002), this cannot have caused the differences in outcome. Third, the pitches of the intervening distractor sequence must be ignored (in order not to overwrite the stored memory of the standard pitch, Deutsch, 1972) whilst at the same time using the temporal information (rhythmicity) of the distractor tones for forming expectations about the occurrence of the comparison tone. According to dynamic attending theory, temporal attention should become more and more specific with each distractor tone (cf. Fig. 1) such that attentional allocation reaches a peak at the expected time-point of the comparison tone, leading to performance benefits. However, recent results show that rhythmicity can also be used to ignore input that is currently irrelevant (Andreou et al., 2011; Devergie et al., 2010; Rimmele et al., 2012). In this case, attentional allocation would reach a trough at the expected time-point of the comparison tone, leading to performance impairments. If the present participants showed a mixture of these strategies (either inter- or intra-individually), this would explain the absence of overall effects. It seems feasible that the participants of Jones et al. (2002) were less inclined to ignore the distractor sequence compared to our participants who might have completely ignored the distractor sequence as instructed, because they might have implicitly realized that the last distractor tone helped them to solve the task by repeating the standard pitch (see Section 1).

Another explanation for the lack of the effect could be that participants were not able to extract the rhythmicity of the distractor sequence at all because they did not pay any attention to the distractor tones, as specifically instructed. However, there is compelling neuroscientific evidence that rhythmicity (especially in its simplest case, isochronicity) can easily be extracted in a bottom-up manner, without any need for attentional allocation (Leung et al., 2013; Schwartze et al., 2012; Takegata and Morotomi, 1999). Therefore, we consider it highly unlikely that the participants did not pick up the rhythm of the distractor sequence in our experiments.

Another alternative explanation is that all participants did use the rhythmicity to reach an attentional peak at the expected (in-phase) time-point of the comparison tone, but that this was nevertheless not beneficial to task performance because optimal attentional allocation was not required for the task. In the task, participants were required to compare the pitch of the standard and comparison tones. Thus, their judgment was based on the frequency rather than the time domain. Dynamic attending theory predicts that the presentation of a regular rhythm entrains attentional oscillations, which in turn creates higher expectations for upcoming events (e.g. sounds) and leads to better discrimination performance. Thus, temporal discrimination thresholds should improve in the context of temporal regularity, if the critical event occurs on time with the previously learned regularity. Indeed, previous research using a time-judgment task preceded by a regular entraining rhythm did show evidence in favor of attentional entrainment (i.e., detrimental effects of shifting the comparison tone out-of-phase; Barnes and Jones, 2000; Boltz, 1993; Jones and Boltz, 1989; Jones, 1976; Large and Jones, 1999; McAuley and Jones, 2003; McAuley and Kidd, 1998; Miller and McAuley, 2005). Further, a recent study by McAuley and Fromboluti (2014) investigated systematic distortions in perceived event duration using an auditory oddball paradigm and found consistent results with the dynamic attending theory. Duration judgments for the rare deviant were less distorted when they referred to events on time with the entrained rhythm, whereas durations of early and late deviants were perceived to be shorter and longer, respectively. Likewise, Lange (2009) found evidence for effects of temporal expectation on judging the presence of a short gap within the stimulus. One important implication arising from our findings is that such judgments of temporal stimulus features lend themselves more readily to the investigation of dynamic attending than judgments of other features such as pitch. When the task requires pitch judgments, it might be too easy to re-orient attention as soon as the comparison stimulus occurs, without losing any task-relevant information by the fact that attention was not optimally allocated in the first place. In contrast, there is evidence in the visual modality that temporal expectations influence visual orientation discrimination and strong interactions between temporal and spatial expectations have been found, which is supported by neurophysiological data (Correa and Nobre, 2008; Doherty et al., 2005; Rohenkohl et al., 2012). However, the stimuli used in the visual task by Rohenkohl et al. (2012) and in similar paradigms were presented very briefly (50 ms). In this case, attending the onset is crucial for extracting temporal information whereas in the pitch-comparison task with 150 ms stimuli, attention could be easily re-oriented if the beginning was missed and thus temporal attention might be less important.

Additionally, auditory timing might special in the context of auditory rhythms. Indeed, research by Prince et al. (2009a, 2009b), trying to disentangle pitch and timing as well as atonal and tonal context within the framework of Jones' et al. (2002) dynamic attending theory, confirms the notion that the comparison of pitch receives attention independently of temporal relations in the current pitch-comparison task. The authors noted an asymmetry between pitch and time relations, with pitch variation influencing temporal processing but temporal variation not influencing pitch processing. The authors concluded that musical context involuntarily invokes greater, and more preferential, attention to pitch, as listeners have learned over years of musical exposure to draw their attention to the dimension of pitch instead of the often non-relevant timing information. In a musical context, pitch is the more elaborate dimension by far, as it contains more variation, uniqueness, and less predictability, and thus may

inherently draw the listener's attention to this dimension at the expense of time (Prince et al., 2009a, 2009b). Previous research has also shown that the processing of pitch-time relations is not influenced by musical expertise (Lebrun-Guillaud and Tillmann, 2007; Makris and Mullet, 2003; Tillmann and Lebrun-Guillaud, 2006). This is consistent with the lack of effects when separating participants by their musicality: The null-results in Experiments 2 to 4 cannot be ascribed to variable musical expertise in our samples.

In addition to the fact that we could not replicate the hypothesized inverted U-shaped expectancy profile, the overall task performance in all five experiments was on average 15% lower than in the original study by Jones et al. (2002). One might therefore argue that our modifications were simply too difficult to produce an effect of entrainment. However, this appears unlikely given that the performance in each of the five experiments, including the exact replication, is well above the chance level of 33%. Further, there is no theoretical reason to constrain the effect to certain individuals (e.g. with less than 6 years of musical training) or within a certain range of task performance (e.g. 70% correct but not 56%; note that both values are well above chance and below ceiling level performance). Likewise, the exact replication did not yield a significant improvement in overall performance. Thus, the assumption by Jones et al. (2002) that the repetition of the standard tone prevents spurious biasing from the pitch difference between the final distractor tone and the comparison tone, thus boosting the accuracy levels, seems less plausible. We argue that repeating the standard tone before the comparison tone fundamentally alters the task, as it becomes effectively a task-relevant tone as stated in the introduction. This may induce strategy differences between participants or even within participants as soon as they discover this particular fact, leading to differences in performance outcomes.

While putting the specific experimental implementation of dynamic attending by the pitch comparison task into perspective, we would like to emphasize that there is a wealth of behavioral and neuroscientific evidence in favor of dynamic attending as such. As outlined above, behavioral benefits in terms of higher accuracies, better detection thresholds, or faster reaction times have been observed for time judgments or other tasks that require temporally precise attention (Barnes and Jones, 2000; Lange and Röder, 2006; Lange, 2009; Large and Jones, 1999; McAuley and Fromboluti, 2014; Rohenkohl et al., 2012). Neural evidence comes from oscillatory entrainment studies (Henry and Obleser, 2012; Lakatos et al., 2013, 2008; Oever et al., 2014; Schroeder and Lakatos, 2009), and underlying mechanisms such as intra-modal and cross-modal phase reset have been proposed (Thorne et al., 2011; Thorne and Debener, 2014). In order to better link sensory entrainment (indicated by behavioral benefits) to neural entrainment (indicated by brain correlates), it is important to continually improve the range of paradigms available for testing both forms of entrainment. This should eventually lead to a better understanding of mechanisms underlying dynamic attending.

Recently, the importance of replication studies has been noted and that the skepticism towards null-results in neuroscience as well as music psychology research should be overcome (Ferguson and Heene, 2012; Frieler et al., 2013; Simmons et al., 2011). Only through replication studies can the growing concern about the reliability and validity of empirical knowledge be confronted (Asendorpf et al., 2013; Nosek et al., 2012). Moreover, replication studies cannot only increase scientific quality management, and serve as a safety net against methodological concerns, or causally influential but unknown factors (Frieler et al., 2013), but can also prevent publication bias in favor of significant results (Ferguson and Heene, 2012). Further, as we failed to replicate the effect in five independent experiments with sufficient power, we conclude that the true effect size must be much lower than the data of Jones et al. (2002) suggest. Since other studies employing different paradigms reported results in favor of attentional entrainment (Barnes and Jones, 2000; Large and Jones, 1999; McAuley and Fromboluti, 2014), we do not question the underlying theory of dynamic attending and the adaption of internal oscillations to external stimuli. However, we question the suitability of the pitch comparison paradigm for the demonstration of auditory dynamic attending. Future paradigms, seeking to investigate sensory as well as neural entrainment, should be modified so that paying attention at the "right moment in time" is more clearly beneficial for task performance than it is for making pitch judgments.

4. Experimental procedures

4.1. *Participants*

In total 140 normal-hearing subjects (73 female) aged 19 to 38 ($M=24.1$, $SD=3.1$; Experiment 1: $N=40$, Experiment 2: $N=23$, Experiment 3: $N=23$, Experiment 4: $N=23$, Experiment 5: $N=31$) participated in the five experiments, no participant more than once, after providing written informed consent. All participants were right-handed and reported no history of diseases affecting the central nervous system. None of the participants reported possessing absolute pitch. In Experiment 1 and 5 each subject had less than 6 years of formal musical training (Experiment 1: $M=2.43$, $SD=1.92$, Experiment 5: $M=1.40$, $SD=1.40$). In Experiments 2 to 4 musicality was assessed with the Goldsmiths Musical Sophistication Index (Gold-MSI; Müllensiefen et al., 2014; German version: Schaal et al., 2014). The data of four participants were discarded due to technical problems (one participant in Experiments 2 and 3, respectively) or inability to perform the pitch comparison task (Experiment 4 and 5). Participants received financial compensation of 8€ per hour. The study was approved by the local ethics committee of the University of Oldenburg.

4.2. *Stimulus material*

Auditory stimuli were generated with Matlab 2012a (Mathworks Inc., Natick, MA) with a 44100 Hz sampling rate and presented binaurally over EAR-Tone 3A insert earphones (3 M Auditory Systems, Indianapolis, United States) at a comfortable listening level in Experiments 1 to 4 and over Sony Professional Headphones (Sony Professional, MDR7506,

San Diego, CA, USA) in Experiment 5. In Experiment 1 and 5 standard and comparison tones had a duration of 150 ms and distractor tones had a duration of 60 ms, as in Jones et al. (2002). In Experiments 2 to 4, the duration of the standard tone was adjusted individually, using lengths from 100 to 250 ms in steps of 25 ms. Further, in Experiment 3 the comparison tone was shortened to 100 ms. In Experiment 4 distractor tones were underlined with a metronome click. Each generated tone had 5 ms onset and offset raised-cosine slopes. All auditory stimuli and the fixation cross were presented and controlled via Presentation software (Version 16.3 10.07.12, Neurobehavioral Systems Inc., Albany, US) and applied using a programmable attenuator (PA5, DT Tucker-Davis Technologies, Indianapolis). Stimuli were adjusted to equal loudness as calibrated with the loudness toolbox (www.genesisacoustics.com) and according to ANSI S3.4-2007 standard.

Stimulus-onset asynchronies (SOA) were 600 ms in Experiment 1 and 5 and were changed to 500 ms in Experiments 2 to 4, as this value is a more characteristic of preferred tempo in music (see also Section 4.4.2). In Experiment 1, the comparison tone occurred equally often at a critical SOA of 524 ms, 600 ms or 676 ms. For Experiments 2 to 4, the critical SOAs were 437 ms, 500 ms, and 563 ms, respectively. In Experiment 5, the critical SOAs were 524 ms, 579 ms, 600 ms, 621 ms, and 676 ms, respectively. For all five experiments standard tones (sine tones) were randomly assigned one of six frequencies (musical pitch values in parentheses): 415 Hz (A-flat$_4$), 440 Hz (A_4), 466 Hz (B-flat$_4$), 622 Hz (E-flat$_5$), 659 Hz (E_5), or 698 Hz (F_5). Equally often the comparison tone was the same pitch as the standard tone or higher or lower by one semitone (each occurring 33% of the time). Intervening distractor tones varied randomly within three semitones centered on 659 Hz if the standard tone was 415 Hz, 440 Hz, or 466 Hz. Correspondingly, distractor tones varied in a range of three semitones and were centered on 440 Hz if the standard tone was 622 Hz, 659 Hz, or 698 Hz.

All five experiments were run in a dimly lit and sound-shielded room, and participants were seated approximately 1.7 m from the screen. During the presentation of sound stimuli participants fixated a white cross on a black background (rgb: 0, 0, 0) in Experiment 1 and a black fixation cross on a gray background (rgb: 125, 125, 125) in Experiments 2 to 5. The fixation cross preceded the standard tone by 500 ms in all five experiments and lasted until the end of the trial. Participants responded by a button press on a standard keyboard and the next trial started after the response was given.

4.3. *Original design*

The general task proposed by Jones et al. (2002) was a pitch comparison task, where the initial standard tone had to be compared to a final comparison tone (see Fig. 1A for illustration). In-between, a series of eight distractor tones of different frequencies (see Section 4.2) was presented at a regular pace. The SOA was 600 ms and five critical SOAs were implemented: 524 ms (very early), 579 ms (early), 600 ms (expected), 621 ms (late), or 676 ms (very late). Subjects were told to ignore the intervening tones and to judge comparison pitches as “higher”, “same” or “lower”. After 15 practice trials with feedback, they received 180 experimental trials, thus 36 trials for each critical SOA, presented in four blocks with no feedback.

4.4. *Modifications and experimental procedure*

To investigate the influence of rhythmic entrainment on task performance and to overcome the methodological concern of the original study by Jones et al. (2002), we modified the original design and subsequently created four different versions, aiming to replicate the initial findings.

4.4.1. *Experiment 1*

Experiment 1 was a combined behavioral and electroencephalographic (EEG) study. Due to the absence of behavioral effects, the EEG data are not reported or interpreted here. Besides the regular SOAs of 600 ms, an irregular condition, presented in different blocks was also implemented. As the irregular condition was implemented to show that non-rhythmic stimuli provide no improvement of task accuracy at (pseudo-)expected time points, the results are not of interest for the current research question. In contrast to the original study, Experiment 1 did not repeat the standard pitch in the last distractor tone, but instead repeated the standard pitch in the first two distractor tones (see Fig. 1B). The repetition of the standard pitch in the first two distractor tones was to better encode the standard pitch as shown in a study by Deutsch (1972). Both duration of standard and comparison (150 ms) as well as duration of the distractors tones (60 ms) were identical to the original study by Jones et al. (2002). Further and in contrast to Jones et al. (2002) only three critical SOAs were implemented (524 ms, 600 ms, 676 ms) because they gave the clearest results in the previous study. Per critical SOA, participants were presented with 72 trials. The order of the regular and irregular conditions was counterbalanced across subjects.

Instructions were identical to the original study (Jones et al., 2002), thus participants were told to ignore the intervening distractor tones and judge the comparison pitch as either “higher”, “same” or “lower” as indicated by a button press on a keyboard. Further, they had to answer as fast and as accurately as possible. Each experiment started with a 15-trial training with feedback. In the main experiment, 216 trials were presented in four blocks with 54 trials each. The blocks were separated by 1-min breaks. During the main experiment, participants received no feedback on their performance. Between the regular and irregular conditions, participants could take a longer break (approximately 5 to 10 min). Each condition, regular and irregular, lasted about 30 min. In the end, participants completed a brief questionnaire on their musical background (years of formal musical training), the difficulty of the task, and any strategies used during the task.

4.4.2. *Experiment 2*

The repetition of the standard pitch as the first two distractors in Experiment 1 might have resulted in better sensory encoding of the standard tone. However, this might have reduced the attentional entrainment effect as better sensory encoding might result in less reliance on contextual

information, which is critical for attentional entrainment. Therefore, the standard tone was not repeated in the intervening distractor sequence in Experiment 2. However, to still ensure adequate encoding of the standard pitch, the duration of the standard tone was adjusted individually by means of a two-down one-up adaptive tracking procedure, aiming for 70.7% accuracy (similar to Levitt, 1971; see Fig. 1C). This individual adjustment was intended to facilitate task performance, without the need to repeat the standard tone. Further, experimental evidence by Van Noorden and Moelants (1999) as well as Moelants (2002) has shown that preferred tempo is located at around 120 beats-per-minute (500 ms) instead of the previously assumed 100 beats-per-minute rate (600 ms; Fraisse, 1982). Thus, the SOA was set to 500 ms instead of 600 ms, and the critical SOAs were changed to 437 ms, 500 ms, and 563 ms accordingly. Finally, the occurrence of the comparison tone had been predictable both in the study by Jones et al. (2002) and in Experiment 1, as the number of distractors was not manipulated (i.e., constant number of 8 distractors). Therefore, we included 25% "catch" trials with a different number of distractors within the 216 trials, which were not further analyzed. This resulted in 18 trials each with 4, 6 or 10 distractors and 54 trials for each critical SOA. This manipulation of varying distractor sequences was intended to increase the overall uncertainty and thus possibly boost the beneficial effect of sensory entrainment in this paradigm. In contrast to the study by Jones et al. (2002), who only tested participants with less than 6 years of formal musical training, participants with diverse musical background were tested in Experiment 2 to investigate the influence of musicality. Musical expertise was assessed with the Goldsmiths Musical Sophistication Index (Gold-MSI; Müllensiefen et al., 2014; German version: Schaal et al., 2014) which uses a multi-facet approach to determine an individual's musical expertise. The Gold-MSI is thereby based on the assumption that musicality is comprised of different facets of musicality, that is, Active Engagement, Singing Abilities, Perceptual Abilities, Musical Training and the Emotional Engagement with Music. Further, a General Musical Sophistication Factor can be derived that determines musical expertise on an interval-based scale.

Participants completed the Gold-MSI before the pitch comparison task. Instructions were identical to Experiment 1, thus participants were asked to judge whether the comparison tone was "higher", "same" or "lower" while ignoring the distractor sequence. The experiment started with a 15 trial training with feedback, using the longest standard tone duration (i.e., 250 ms). This was followed by the adaptive tracking procedure. On each trial, participants had to judge the pitch as "higher", "same" or "lower", as in the training and main experiment. Listeners completed one block of the adaptive tracking procedure containing 25 trials in about 5 min. Individual threshold was determined as the arithmetic average of the last 5 trials. In order not to bias the adaptive tracking procedure, only trials with expected SOA were presented and participants received no feedback. Subsequently a second training, identical to the first training, was completed with a subject's adjusted duration of the standard tone. In the main experiment 216 trials were presented in four blocks with 54 trials each. Each block was followed by a one-minute break.

4.4.3. *Experiment 3*

The individual adaptation of the standard tone duration, the SOA of 500 ms as well as the varying distractor sequence lengths with 25% occurrence of 4, 6 or 10 intervening distractors were adopted from Experiment 2. Because the occurrence of the comparison tone was the crucial manipulation in the current paradigm, the main aim of Experiment 3 was to focus on the comparison tone rather than the standard tone. The comparison tone was shortened to 100 ms instead of 150 ms (see Fig. 1D). The rationale for shortening the comparison tone was to avoid opportunities for attentional re-orienting.

Setup, procedure, and instructions were the same as in Experiment 2.

4.4.4. *Experiment 4*

The individual adaptation of the standard tone duration, the SOA of 500 ms and 25% catch trials with varying distractor sequence lengths were adopted from Experiments 2 and 3. Duration of the comparison tone was set back to 150 ms. A potential drawback identified in Experiments 1 to 3 was that the distractor tones might interfere too strongly with the pitch comparison (Deutsch, 1972). To reduce this interference and to provide at the same time a clear onset and thus a successful sensory entrainment, distractor tones were accompanied by metronome clicks in Experiment 4, which provides additional pitch-independent temporal information (for further details see Prince et al., 2009b). It has previously been shown that auditory sequences even without beat-inducing cues result in beat perception, a phenomenon known as "subjective rhythmization" (e.g. Brochard et al., 2003; Fraisse, 1982). A common manifestation of subjective rhythmization is the so-called clock illusion, when a regular sound ('tick-tick-tick-tick…') is spontaneously grouped into a binary pattern ('tick-tock-tick-tock…'). In Experiments 1 to 3 subjective accentuation might have already started with the standard tone, resulting in a subjective unaccented time point for the comparison tone. To ensure that the comparison tone was at a subjectively accented time-point we included a silent 500 ms cycle between the standard tone and the first distractor. For an illustration see Fig. 1E.

Setup, procedure and instructions were otherwise the same as in Experiments 2 and 3.

4.4.5. *Experiment 5*

In Experiment 5 we implemented an exact replication of the original experiment by Jones et al. (2002; see Section 4.3 for details) and also included a second condition "without repetition", where the standard tone was not repeated as the last distractor tone. We used a within-subject design in which the order of the two conditions (with and without repetition) was counter-balanced over participants. The length of the standard and comparison tone was set back to 150 ms. Further, the silent cycle and metronome were excluded and all distractor sequences had the same length, thus 8 distractors

each. Only participants with less than 6 years of formal musical training were invited for the experiment. After 15 practice trials with feedback, they received 180 experimental trials, thus 36 trials for each critical SOA (524 ms (very early), 579 ms (early), 600 ms (expected), 621 ms (late), or 676 ms (very late)), presented in four blocks with no feedback and with intervening breaks of 1-min. After each experimental condition participants responded to a brief questionnaire on the task and the strategies used. Instructions were the same as in the Experiments 1 to 4.

4.5. *Power analysis*

To avoid the risk of false negative findings due to underpowered sample size (Ellis, 2010) and to be able to interpret null findings (Aberson, 2002), an a priori power analysis was conducted for Experiments 2 to 4. The original study by Jones et al. (2002) used a within-subject design with accuracy as dependent variable. Following the first experiment, sample size was calculated for Experiments 2 to 4 by conducting an a priori analysis with G*Power V 3.1.9.2 software (Faul et al., 2007), using the option "ANOVA: Repeated measures, within factors". We determined an approximate effect size using the statistical information in Jones et al. (2002) yielding a $\eta^2=.16$, thus a large effect size and a power of $1-\beta=.99$. As this calculation takes the manipulation of 5 critical SOAs into account, we subsequently performed a second power analysis using the mean accuracies of the critical SOAs. Using the option "Means: Difference between two dependent means (matched pairs)" we determined an approximate effect size yielding a Cohen's $d_z=.87$ and a power of $1-\beta=.99$, thus also a large effect size. However, as overestimation of the true effect sizes is often problematic (Button et al., 2013), we decided to use a range of effect sizes with a constant power of $1-\beta=.80$ ($\alpha=.05$) as a plausible value (Cohen, 1988). A small effect size of $\eta^2=.01$ would result in a group size of approximately $N=161$ and a large effect size of $\eta^2=.14$ in a group size of approximately $N=12$. As we estimated the true effect to be between medium and large, we ultimately decided to set the required group size for each experiment to $N=23$, which allowed us to observe a minimum effect size of approximately $\eta^2=.07$.

4.6. *Statistical analysis*

All statistical analyses were performed with the R programming language for statistical computing (version 3.0.1, R Core Team, Vienna, Austria). To test for effects of the temporal manipulation on performance, a one-way repeated-measures ANOVA with the factor *timing of comparison tone* (3 levels: very early/on time/very late) was performed separately for Experiments 1 to 4 with accuracy as the dependent variable. To test for effects of temporal manipulation in Experiment 5 a 2×5 repeated-measures ANOVA with factors *timing of comparison tone* (5 levels: very early/early/on time/late/very late) and *condition* (2 levels: with repetition/without repetition) was performed with accuracy as the dependent variable. In order to test accuracy improvements between training runs 1 and 2 (i.e., before and after applying the adaptive procedure for the duration of the standard tone), a paired-sample two-tailed Student's *t*-test was used. A Pearson product-moment correlation was calculated to test the influence of musicality on task performance. Further, a global Pearson product-moment correlation between musicality and task performance was conducted across all individuals (Experiments 2 to 4), following *z*-transformation to account for differences in task difficulty between experiments. A median split on the General Musical Sophistication Factor categorized participants into musicians and non-musicians separately for each experiment. A post-hoc independent-sample, two-tailed Student's *t*-test was used to test differences in the influence of rhythmicity (i.e., on-time presentation vs. early or late presentation) between musicians and non-musicians. Assumptions of the statistical tests were evaluated using the respective test's methods. The significance level was set to $p<.05$ for all tests and correlations.

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